Ex no: 9

Date: 22.8.26

Implementation of subnetting in CISCO PACKET TRACER Simulator.

a) Design Multiple subnets with a suitable Number of Host.

Aim:

To Design Multiple subnets with a suitable number of hosts using subnetting technique in CISCO Packet Tracer.

Steps:

1) Open CISCO Packet Tracer and create a new project.
2) Design the network topology by planning the required network devices.
3) Select a suitable network topology for the experiment.
4) Calculate the number of subnets required.
5) Determine the no of host needed in each subnet.
6) Calculate the subnet mask using subnetting techniques.
7) Divide the network into multiple subnets.
8) Assign a seperate network address to each subnet.
9) Verify the subnet calculations and host ranges.

Result:

Thus, Multiple subnets were sucessfully designed according to the requirements of hosts

b) Assign static IP Address Across all subnets and connect the subnet through a Router.

Aim:

To assign static IP address to all devices in different subnet and establish communication between the subnet through a router.

Steps:

1) Connect a router in the switches using appropriate cable.
2) Configure the router interface.
3) Assign an IP address and subnet mask to end other router.
4) Configure the switches.
5) Assign an IP address to all computer.
6) Configure the subnet mask for each done.
7) Assign the appropriate defaults gateway to each computer.
8) Verify the IP configuration of all devices.
9) Save the router configurations.
10) Verify the all devices are properly connected.

Result:

Thus, static IP address were assigned sucessfully and communication between the subnets was established through the router

c) Simulate Packet Transmission across the subnets and observe the result:

a) when subnets are connected through a Router.

Aim:

To Analyse Packet Transmission between different subnet when a router is used for communication.

Steps:

1) Open simulator mode in Cisco Packet Tracer.
2) Select a source device from one subnet.
3) Select a destination device from another subnet.
4) Generate a packet using the simple Protocol.
5) Observe the movement of the packet through the network.
6) Analyse how the router receive the Packet.
7) Observe the forwarding process performed by the router.
8) Verify the Packet information between different subnet.
9) Use the ping command to test connectivity.
10) Observe the successfully delivery of packets.
11) Record the simulation results.
12) Analyse the Router process.

Result:

Thus, Packets were transmitted successfully between different subnet through the router

b) when subnets are not connected through a Router.

Aim:
To Analyse Packet transmission between different subnets when a router is not used for communication.

Steps:
1) Remove the router from the network topology.
2) Open simulation mode.
3) Select device belonging to different subnets.
4) Generate packets using the simple protocol.
5) Observe the packet transmission.
6) Perform a ping test between devices located different subnets.
7) Analyse the Packet flow.
8) Observe the simulation events.
9) Identify the points at which packet transmission fails
10) Compare the results with the network that use a router.
11) Record the observations
12) Analyse the implemented of a router in inter network communication

Result:
Thus, communication between [illegible] because a router was used available to [illegible]

PC0

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.11

Pinging 192.168.3.11 with 32 bytes of data:

Request timed out.
Reply from 192.168.3.11: bytes=32 time<1ms TTL=126
Reply from 192.168.3.11: bytes=32 time<1ms TTL=126
Reply from 192.168.3.11: bytes=32 time=11ms TTL=126

Ping statistics for 192.168.3.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 3ms

C:\>ping 192.168.4.11

Pinging 192.168.4.11 with 32 bytes of data:

Request timed out.
Reply from 192.168.4.11: bytes=32 time=10ms TTL=126
Reply from 192.168.4.11: bytes=32 time<1ms TTL=126
Reply from 192.168.4.11: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.4.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>
```

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip address 192.168.3.0 255.255.255.224 10.0.0.2
                  ^
% Invalid input detected at '^' marker.

Router(config)#ip route 192.168.3.0 255.255.255.224 10.0.0.2
Router(config)#ip route 192.168.4.0 255.255.255.224 10.0.0.2
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#enable
Router#ping 10.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/11/46 ms

Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     192.168.2.1     YES manual up                    up
GigabitEthernet0/2     10.0.0.1        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 192.168.1.0 255.255.255.224 10.0.0.1
Router(config)#ip route 192.168.2.0 255.255.255.224 10.0.0.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#enable
Router#ping 10.0.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.3.1     YES manual up                    up
GigabitEthernet0/1     192.168.4.1     YES manual up                    up
GigabitEthernet0/2     10.0.0.2        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
Router#
```

Router1

Physical Config CLI Attributes

IOS Command Line Int

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.224
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.224
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#interface GigabitEthernet0/2
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
```

Router2

Physical Config CLI Attributes

IOS Command Lin

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.3.1 255.255.255.224
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 192.168.4.1 255.255.255.224
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#interface GigabitEthernet0/2
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

